

# Relation Algebra

## §5. Datatypes in Allegories

### §5.1. Relators

(5.1a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from  $R$  to  $S$  IS  $R \subseteq S$ . A **relator**  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \subseteq S \implies F(R) \subseteq F(S)$$

Preserving  $\circ$  is **not** asked for —  $\circ$  is an identity-on-objects involution  $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$ , no part of the 2-category.

(5.1b)

**the statement**

$\text{Map}(f) \implies \text{Map}(F(f))$  and  $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$ .

Over a **tabular** allegory a functor is a relator  $\iff$  it preserves  $\circ$ .

$F(R^\circ) = F(R)^\circ$  for every  $R$ , so  $F(R)^\circ$  needs no bracket.

Two relators agreeing on maps are equal.

$\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$

$F(R \cap S) \subseteq F(R) \cap F(S)$ , and strictly.

$\text{PreservesRecip}(F) \implies F(\text{dom}(R)) = \text{dom}(F(R))$

The **power relator**  $P : \mathcal{C} \rightarrow \mathcal{C} : P(R) = \{ (a, b) \mid \exists x \in \text{dom}(R). a R x \wedge \exists y \in \text{cod}(R). x S y \}$  — is where the fourth is strict: for  $R = \{(a_1, b_1), (a_2, b_2)\}$  and  $S = \{(a_1, b_2), (a_2, b_1)\}$  the pair  $(\{a_1, a_2\}, \{b_1, b_2\})$  is in  $P(R) \cap P(S)$ , while  $R \cap S = \emptyset$ .

### §5.2. Fork $\langle R, S \rangle$

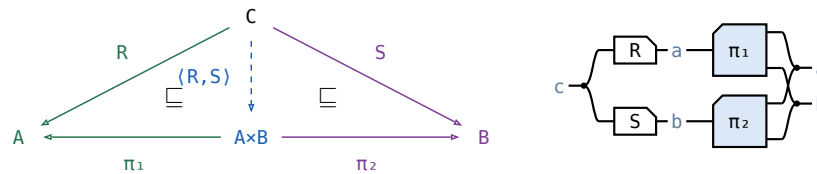
(5.2a)

The **fork** of  $R : \mathcal{C} \rightarrow \mathcal{A}$  and  $S : \mathcal{C} \rightarrow \mathcal{B}$  is  $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$ , where  $(\pi_1, \pi_2)$  is the tabulation of  $\tau$ .

(5.2b)

$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad \langle R, S \rangle \pi_2 = \text{dom}(R) S$$

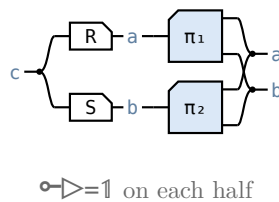
(5.2c)



A domain is coreflexive, so  $\langle R, S \rangle \pi_1 \subseteq R$ , with equality exactly when  $S$  is entire; for maps both triangles commute and  $\langle f, g \rangle$  is unique. In  $\text{Rel}$ ,  $c \in \langle R, S \rangle$  iff  $c R a$  and  $c S b$  — copy  $c$ , then  $R$  on one strand and  $S$  on the other, which is  $\triangleleft (R \otimes S)$  on the right.

No  $\circ$  survives the translation.  $\pi_1 = 1 \otimes \circ$  discards the second component, so  $\pi_1^\circ = 1 \otimes \bullet$  **creates** one out of nothing, and  $\cap$  is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(5.2d)

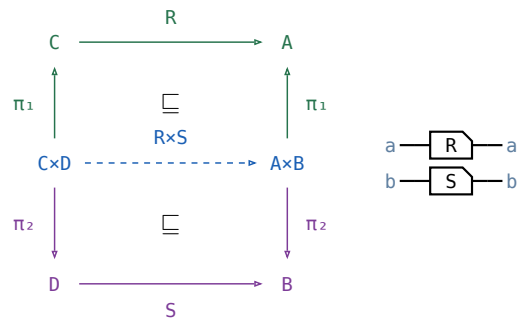


#### §5.2.1. Relational product $R \times S$

(5.2.1a)

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$ , a relator in each argument but no longer a categorical product.

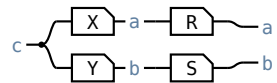
(5.2.1b)



Right-then-up is  $(R \times S)\pi_1$ , up-then-right is  $\pi_1 R$ , and  $(R \times S)\pi_1 \sqsubseteq \pi_1 R$ , equality when  $S$  is entire. In  $\mathbf{Rel}$ ,  $(c, d) (R \times S) (a, b)$  iff  $c R a$  and  $d S b$  — two strands side by side, no copy dot:  $R \times S = R \otimes S$ .

### §5.2.2. Absorption

For  $X : E \rightarrow C$  and  $Y : E \rightarrow D$ ,  $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ . Both sides are this picture:



(5.2.2a)

(5.2.2b)

| the law                                                                                                                                                                                                                                                                                                                                                            | picture |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 $R \times S \triangleq (\pi_1 R, \pi_2 S)$                                                                                                                                                                                                                                                                                                                       |         |
| 2 $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$<br>both sides are the same strokes — absorption. Its $\sqsubseteq$ half is that half at $S := \mathbb{1}$ and at $R := \mathbb{1}$ — stages of their proof of this row; the corollary is the $(R \times S) (U \times V) = (RU) \times (SV)$ it yields, at $R := \mathbb{1}$ and $V := \mathbb{1}$ . |         |
| 3 $\langle R, S \rangle \pi_1 = \text{dom}(S) R$<br>(5.2b)                                                                                                                                                                                                                                                                                                         |         |
| 4 $\langle R, S \rangle \pi_2 = \text{dom}(R) S$<br>(5.2b)                                                                                                                                                                                                                                                                                                         |         |
| 5 $\langle X, Y \rangle (R, S)^\circ = XR^\circ \cap YS^\circ$                                                                                                                                                                                                                                                                                                     |         |
| 6 $\langle R, S \rangle^\circ \langle U, V \rangle \sqsubseteq (R^\circ U) \times (S^\circ V)$                                                                                                                                                                                                                                                                     |         |
| 7 $\text{Map}(f) \implies f \langle R, S \rangle = \langle fR, fS \rangle$<br>$f$ a map; it fails for an arbitrary arrow;                                                                                                                                                                                                                                          |         |
| 8 $F(R \times S) \text{unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$<br>$\text{unzip}(F) \triangleq (F(\pi_1), F(\pi_2))$ ;<br>only $\sqsubseteq$ for an arbitrary $R, S$                                                                                                                                                                              |         |
| 9 $g = \text{curry}(f) \iff (g \times \mathbb{1}) \text{eval} = f$<br>reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?                                                                                                                                                                                                           |         |

### §5.3. Coproduct $[R, S] : A + B \rightarrow C$

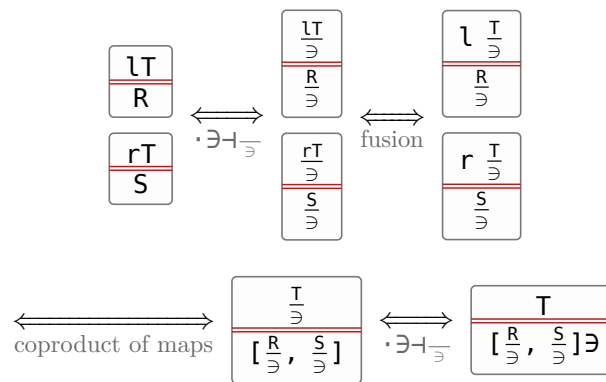
(5.3a)

| the statement                                                                                                                                                                                         | picture |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $[R, S] \triangleq l \circ R \cup r \circ S$<br>The tape is the union — a particle entering at <b>A+B</b> takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint. |         |
| $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$                                                                                                                                             |         |
| $R + S \triangleq [Rl, Sr]$                                                                                                                                                                           |         |
| $l[R, S] = R, r[R, S] = S$<br>$lT = R \wedge rT = S \implies T = [R, S]$                                                                                                                              |         |
| $ll^\circ = 1, rr^\circ = 1$                                                                                                                                                                          |         |
| $lr^\circ = 0, rl^\circ = 0$                                                                                                                                                                          |         |
| $l^\circ l \cup r^\circ r = 1$                                                                                                                                                                        |         |
| $[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$                                                                                                                                                      |         |

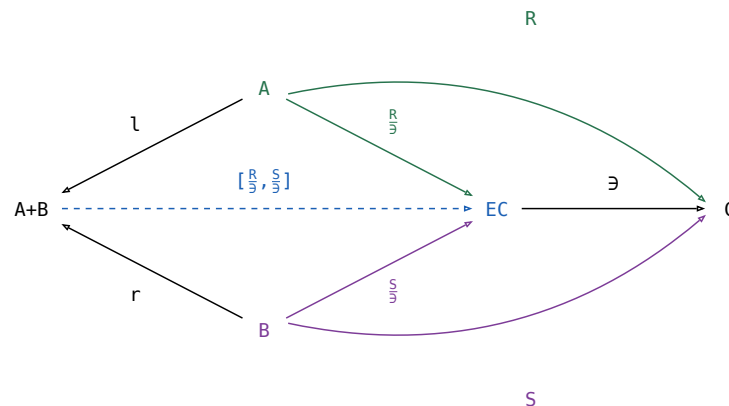
### §5.3.1. $[R, S] \triangleq [\frac{R}{\exists}, \frac{S}{\exists}] \exists$

The universal-property row is not free:  $l, r$  were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because  $\exists$  sends an arrow  $A \rightarrow C$  to a map  $A \rightarrow EC$  reversibly, so the map coproduct can be applied underneath it. For any  $T : A+B \rightarrow C$ ,

(5.3.1a)



(5.3.1b)



Nothing here holds only up to  $\Xi$ : every triangle commutes on the nose, which is the difference from the fork above. The border spells  $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$ , and pushing  $\exists$  into the union that  $[\cdot, \cdot]$  on maps already is turns that back into the definition,  $(\frac{R}{\exists} \cup \frac{S}{\exists}) \exists = \frac{R}{\exists} \exists \cup \frac{S}{\exists} \exists = \frac{R}{\exists} \exists \cup \frac{S}{\exists} \exists$ .

(5.3.1c)

| the law                                                                                                                                                                                                                       | picture |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $[R, S] \triangleq \frac{R}{\exists} \cup \frac{S}{\exists}$<br>(5.3a)'s first row                                                                                                                                            |         |
| $R + S \triangleq [Rl, Sr]$                                                                                                                                                                                                   |         |
| $[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$<br>(5.3a)'s last row;                                                                                                                                                        |         |
| $[1, 0] = \frac{l}{\exists}, [0, 1] = \frac{r}{\exists}$<br>$[l, r] = 1$<br>which is (5.9)                                                                                                                                    |         |
| prove (5.11), and say why duality<br>does not carry it over from the<br>product law                                                                                                                                           |         |
| $(R + S) \cap [P, Q] [U, V]^\circ =$<br>$(R \cap P U^\circ) + (S \cap Q V^\circ)$<br>$[P, Q] [U, V]^\circ$ is a full $2 \times 2$ of<br>composites; $R + S$ is diagonal, so the<br>meet cuts the two off-diagonal<br>branches |         |

## §5.4. The power relator $P(R)$

(5.4a)

For  $R : A \rightarrow B$ ,

$$P(R) \triangleq (\epsilon \setminus R \epsilon) \cap (\exists R / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^\circ$$

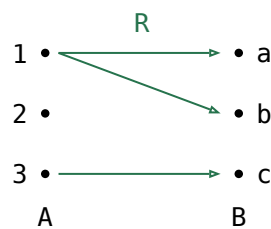
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of  $xs$  is related by  $R$  to some element of  $ys$ , and conversely.

(5.4b)

|                                       | in Rel                                                                                              | in words                                                                        | $\text{im}(xs)$              | partner                                       |
|---------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------|-----------------------------------------------|
| $(\exists R) / \exists$               | $\forall y \in ys. \exists x \in xs. x \ R \ y$                                                     | $\forall y. \text{ some } xs \ R \ y$                                           | $ys \subseteq \text{im}(xs)$ | every $y$ has a partner                       |
| $(\exists / (\exists R))^\circ$       | $\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$                                     | every $y$ with some $xs \ R \ y$ is in $ys$                                     | $ys \supseteq \text{im}(xs)$ | $ys$ leaves out no partner                    |
| $E(R)$                                | $ys = \{y \mid \exists x \in xs. x \ R \ y\}$                                                       | $ys = \text{ every } y \text{ with some } xs \ R \ y$                           | $ys = \text{im}(xs)$         | every $y$ has a partner, and none is left out |
| $((\exists R^\circ) / \exists)^\circ$ | $\forall x \in xs. \exists y \in ys. x \ R \ y$                                                     | $\forall x. x \ R \ \text{ some } ys$                                           | —                            | every $x$ has a partner                       |
| $P(R)$                                | $\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$ | $\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{ some } ys$ | —                            | every $x$ and every $y$ has a partner         |

(5.4c)



| $xs$          | $(\exists R) / \exists$ | $(\exists / (\exists R))^\circ$ | $E(R)$      | $((\exists R^\circ) / \exists)^\circ$ | $P(R)$        |
|---------------|-------------------------|---------------------------------|-------------|---------------------------------------|---------------|
| $\emptyset$   | $\emptyset$             | all 8 subsets of $abc$          | $\emptyset$ | all 8 subsets of $abc$                | $\emptyset$   |
| $\{1\}$       | $\emptyset, a, b, ab$   | $ab, abc$                       | $ab$        | $a, b, ab, ac, bc, abc$               | $a, b, ab$    |
| $\{2\}$       | $\emptyset$             | all 8 subsets of $abc$          | $\emptyset$ | none                                  | none          |
| $\{3\}$       | $\emptyset, c$          | $c, ac, bc, abc$                | $c$         | $c, ac, bc, abc$                      | $c$           |
| $\{1, 2\}$    | $\emptyset, a, b, ab$   | $ab, abc$                       | $ab$        | none                                  | none          |
| $\{1, 3\}$    | all 8 subsets of $abc$  | $abc$                           | $abc$       | $ac, bc, abc$                         | $ac, bc, abc$ |
| $\{2, 3\}$    | $\emptyset, c$          | $c, ac, bc, abc$                | $c$         | none                                  | none          |
| $\{1, 2, 3\}$ | all 8 subsets of $abc$  | $abc$                           | $abc$       | none                                  | none          |

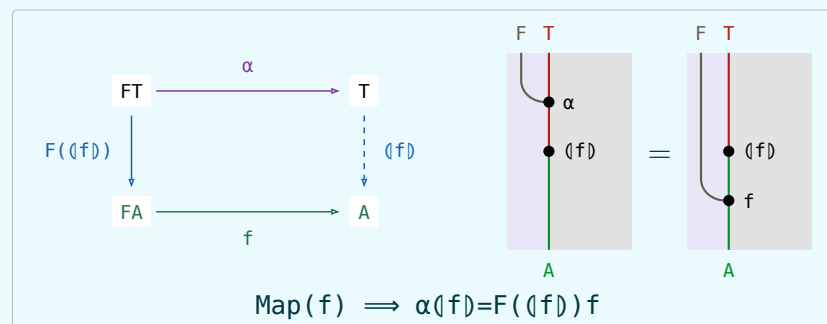
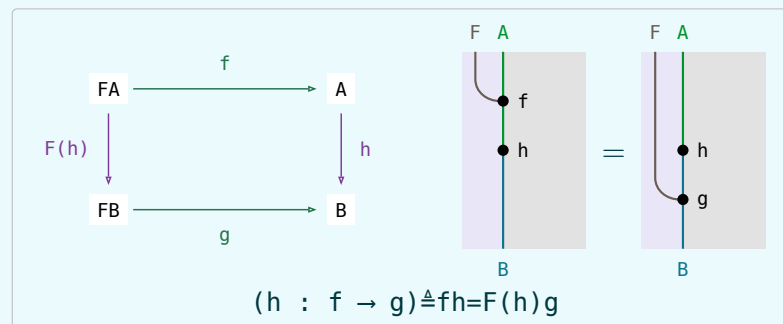
(5.4d)

| law                                                                                                | the reading                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \sqsubseteq P(R) \iff X \sqsubseteq \exists R \text{ and } X^\circ \sqsubseteq \exists R^\circ$ | One containment, and the same one at $R^\circ$ — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$ , and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$ .                                                                                                                  |
| $P(1) = 1$                                                                                         | The straightness axiom verbatim: extensionality is $P$ 's unit law.                                                                                                                                                                                                                                            |
| $\text{Map}(f) \implies P(f) = E(f)$                                                               | In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$ . The half at $f^\circ$ says every $a \in xs$ has its $f(a)$ on $ys$ ; $f$ has just the one image per $a$ , so that already says $ys$ contains everything $xs$ reaches, which is the fraction's second half. For a map the two definitions coincide. |
| $P(RS) = P(R)P(S)$                                                                                 | $\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$ .                                                                                                                                                    |

## §5.5. Initial algebra

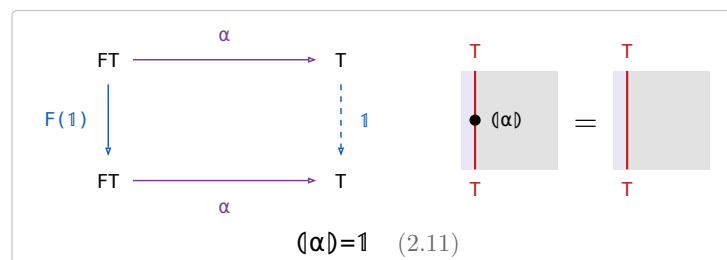
(5.5a)

An **F-algebra** is a map  $f : F(A) \rightarrow A$ ;  $A$  is its **carrier**. An **F-homomorphism** from  $f : F(A) \rightarrow A$  to  $g : F(B) \rightarrow B$  is a map  $h : A \rightarrow B$  with  $f \circ h = F(h) \circ g$ . The **initial algebra**  $\alpha : F(T) \rightarrow T$  is the F-algebra with exactly one F-homomorphism  $\langle f \rangle : T \rightarrow A$  to every F-algebra  $f$ .



### §5.5.1. Reflection

(5.5.1a)

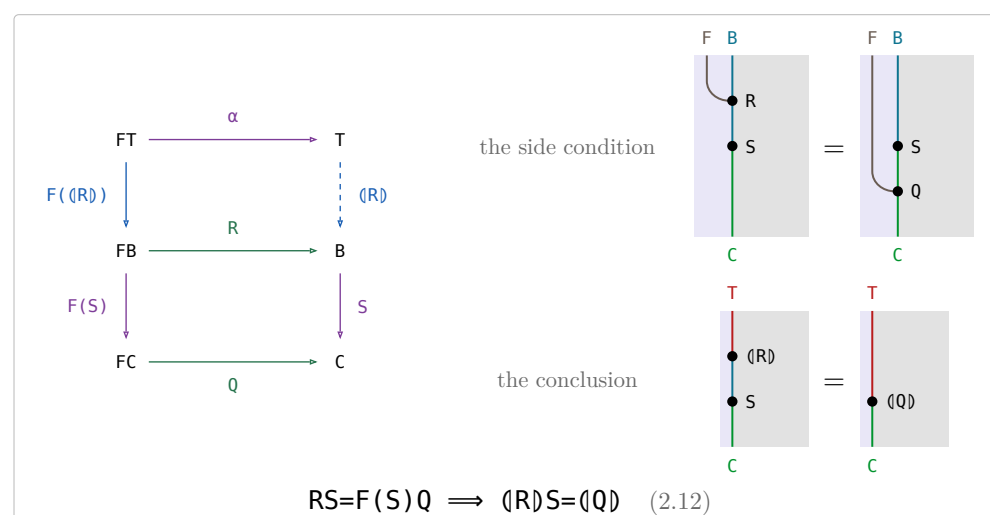


Taking a value apart with  $\alpha$  and putting it straight back is doing nothing.

### §5.5.2. Fusion: $\langle R : FB \rangle$ absorb $S : R \rightarrow Q$ (in $\text{Alg}(F)$ ) and becomes $\langle Q \rangle$

When  $S : B \rightarrow C$  is an F-homomorphism from  $R : F(B) \rightarrow B$  to  $Q : F(C) \rightarrow C$ , folding with  $R$  and then applying  $S$  is folding with  $Q$ .

(5.5.2a)



## §5.6. Type relator

(5.6a)

Let  $F$  be a binary relator with initial type  $(\alpha, T)$ , so  $T$  is a type functor.  $F(R, S)$  is its action on a pair, and  $F(X)$  abbreviates  $F(1, X)$ , the  $F$  of the reduce section. For every object  $A$  the initial algebra is  $\alpha : F(A, TA) \rightarrow TA$ , among the maps.  $T$  acts on an arrow  $R : A \rightarrow B$  by

$$T(R) = (F(R, 1) \alpha)$$

(5.6b)

| name                   | law                               | what it says                                                                                                                                                                                                |
|------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the defining equation  | $T(R) = (F(R, 1) \alpha)$         | Rebuild the structure with $\alpha$ , applying $R$ to the parameter on the way.                                                                                                                             |
| functor                | $T(1) = 1$ and $T(R)T(S) = T(RS)$ | Acting by the identity changes nothing, and two actions in a row are one action.                                                                                                                            |
| type functor fusion    | $T(f)(h) = (F(f, 1)h)$            | A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because $F$ is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$ . |
| naturality of $\alpha$ | $\alpha T(f) = F(f, T(f)) \alpha$ | Building and then mapping is the same as mapping the parts and then building, so $\alpha$ is natural from $G(R) = F(R, T(R))$ to $T$ .                                                                      |
| type relator           | $T(R)^\circ = T(R^\circ)$         | A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.                                                                                                        |

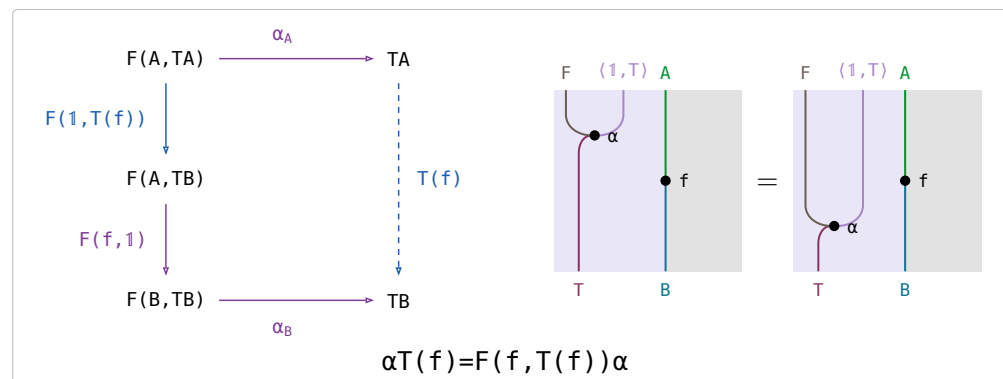
## §5.6.1. Type functor

(5.6.1a)

Let  $F$  be a bifunctor taking both the parameter  $A$  and the recursive position  $TA$ , with an initial algebra  $\alpha_A : F(A, TA) \rightarrow TA$  for every object  $A$ . Then  $T$  is a functor, acting on a map  $f : A \rightarrow B$  by

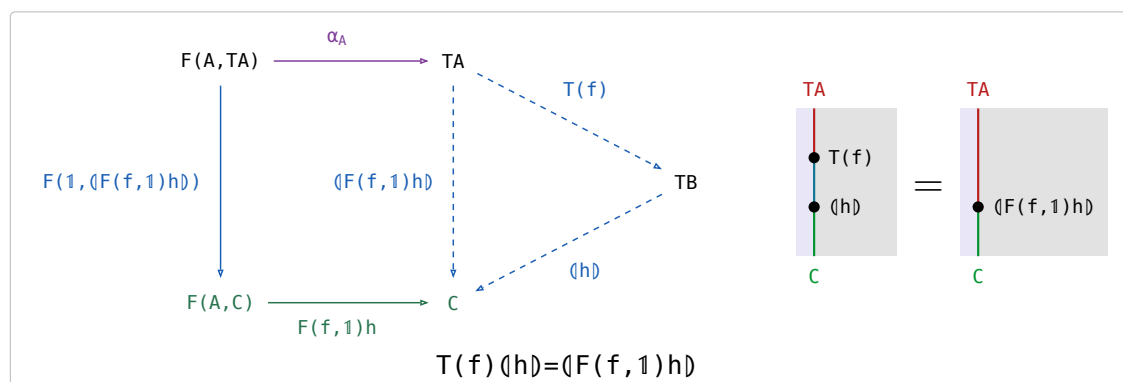
$$T(R) \triangleq (F(R, 1) \alpha)$$

(5.6.1b)



- $F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$  is a bifunctor and a wire is a unary functor, so the two arguments are packed first:  $\langle 1, T \rangle : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$  sends  $A$  to  $(A, TA)$ , and  $F(\langle 1, T \rangle(A))$  is  $F(A, TA)$ .
- The picture is three wires —  $F$ ,  $\langle 1, T \rangle$ , and the object — and the region between the first two is  $\mathcal{A} \times \mathcal{A}$ .
- $\alpha$  is then an ordinary natural transformation  $F \circ \langle 1, T \rangle \Rightarrow T$ : its bead eats the  $F$  and  $\langle 1, T \rangle$  wires, and the  $T$  wire is born under it.
- $F(f, T(f))$  costs no notation. It is the bead  $f$  on the object wire with  $\langle 1, T \rangle$  and  $F$  running past:  $\langle 1, T \rangle$  is what turns  $f$  into the pair  $(f, T(f))$ , and  $F$  is what applies it.
- The law is the naturality of  $\alpha$ , which is exactly the freedom to slide that  $f$  bead past it.

(5.6.1c)



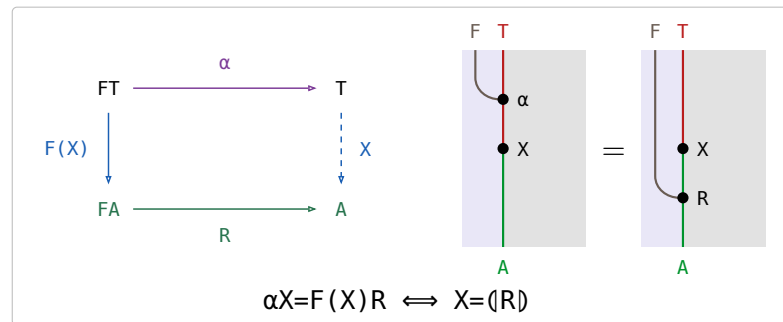
## §5.7. Reduce

(5.7a)

let  $F$  be a relator and has **initial algebra**  $\alpha : F(T) \rightarrow T$  in the subcategory of functions.  $\alpha$  is also initial in the allegory:

### §5.7.1. The defining equation

(5.7.1a)



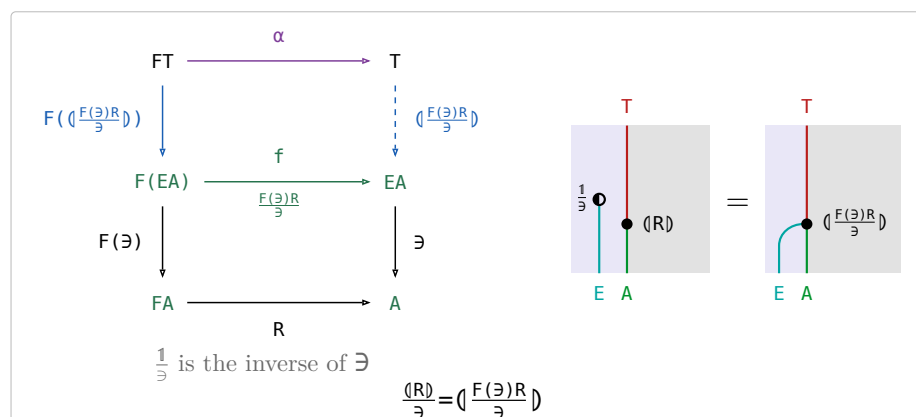
An  $F$ -algebra is a **weakened** natural transformation. A transformation  $F \Rightarrow 1$  on  $\mathcal{A}$  would need a component  $FX \rightarrow X$  at every object and a commuting square at every arrow, but  $F$ -Algebra only need it works on  $T$  and  $A$ .

(5.7.1b)

| $\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$                                                                                                                     |                                                                                                                                        |                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| catamorphism: arrows of the form $\llbracket f \rrbracket$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing |                                                                                                                                        |                                                                                                                                             |
| part                                                                                                                                                                     | Nat                                                                                                                                    | [A]                                                                                                                                         |
| datatype                                                                                                                                                                 | $\text{Nat} ::= \text{zero} \mid \text{succ Nat}$                                                                                      | $[A] ::= \text{nil} \mid \text{cons}(A, [A])$                                                                                               |
| base functor $F$                                                                                                                                                         | $F(X) = 1 + X$                                                                                                                         | $F(X) = 1 + A \times X$                                                                                                                     |
| initial algebra $\alpha$                                                                                                                                                 | $\alpha = [\text{zero}, \text{succ}]$<br>$: 1 + \text{Nat} \rightarrow \text{Nat}$                                                     | $\alpha = [\text{nil}, \text{cons}]$<br>$: 1 + A \times [A] \rightarrow [A]$                                                                |
| the fold, pointwise                                                                                                                                                      | $\llbracket [c, f] \rrbracket(\text{zero}) = c$<br>$\llbracket [c, f] \rrbracket(\text{succ}(n)) = f(\llbracket [c, f] \rrbracket(n))$ | $\llbracket [c, f] \rrbracket(\text{nil}) = c$<br>$\llbracket [c, f] \rrbracket(\text{cons}(a, x)) = f(a, \llbracket [c, f] \rrbracket(x))$ |

### §5.7.2. $\llbracket R \rrbracket = \llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists$

(5.7.2a)



(5.7.2b)

$$\begin{array}{c}
 \boxed{\frac{\alpha X}{F(X)R}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}}} \\
 \xleftrightarrow{\text{relator, fusion twice}} \boxed{\frac{\frac{\alpha}{\frac{X}{\exists}}}{\frac{F(\frac{X}{\exists})}{\frac{F(\exists)R}{\exists}}}} \xleftrightarrow{\text{reduce of maps}} \boxed{\frac{\frac{X}{\exists}}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{X}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists}}
 \end{array}$$

the last two rows at  $X := \llbracket R \rrbracket$ :

(5.7.2c)

$\alpha X = F(X)R \iff \frac{X}{\exists} = (\frac{F(\exists)R}{\exists})$

in a tabular allegory, with  $F$  a relator and  $\alpha : F(T) \rightarrow T$  its initial algebra: a relation  $X$  out of  $T$  satisfies the fold equation  $\alpha X = F(X)R$  of  $R$  exactly when  $\frac{X}{\exists}$  is the fold of the map  $\frac{F(\exists)R}{\exists}$

(a)  $\alpha$  iso;  $\alpha X = F(X)R$   
 (b)  $\alpha^0$  is a map because  $\alpha$  is an iso, so it leaves  $\Lambda$ :  $\frac{\alpha^0 F(X)R}{\exists} = \alpha^0 \frac{F(X)R}{\exists}$  — pow-laws  
 (c)  $X = \frac{X}{\exists}$  — pow-laws  
 (d)  $F(\frac{X}{\exists}) = F(\frac{X}{\exists})F(\exists)$  as  $F$  is a functor, and  $F(\frac{X}{\exists})$  is a map (a relator sends maps to maps), so it leaves  $\Lambda$  —  $\triangleq$  relator, pow-laws  
 (e) fold uniqueness,  $\frac{F(\exists)R}{\exists}$  being a map — (5.5a)

## §5.8. Combinatorial functions

(5.8a)

| definition                                                                                                                        | type                                  | note                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $[A] ::= \text{nil} \mid \text{cons}(A, [A])$                                                                                     | $\mathcal{A} \rightarrow \mathcal{A}$ | The list type, under the short name it keeps.                                                                                                                     |
| $\text{list}(R) = ([\text{nil}, (R \times 1) \text{cons}])$                                                                       | $[A] \rightarrow [B]$                 | The relator's action on $R : A \rightarrow B$ : one $R$ per element, the shape untouched.                                                                         |
| $\text{subseq} = ([\text{nil}, \text{cons } u \pi_2])$                                                                            | $[A] \rightarrow [A]$                 | $xs \text{ subseq } ys$ : $ys$ is $xs$ with elements dropped — $\text{cons}$ keeps the head, $\pi_2$ drops it.                                                    |
| $\text{prefix} = ([\text{nil}, \multimap \text{nil } u \text{cons}])$<br>$\text{prefix} = \text{cat} \circ \pi_1 = \text{init}^*$ | $[A] \rightarrow [A]$                 | $ys$ is an initial segment of $xs$ ; the first $\text{nil}$ is where it stops early.<br>$\text{init} \triangleq \text{snoc} \circ \pi_1$ .                        |
| $\text{suffix} = \text{cat} \circ \pi_2 = \text{tail}^*$                                                                          | $[A] \rightarrow [A]$                 | The dual, $\text{tail} \triangleq \text{cons} \circ \pi_2$ ; as a reduce it needs $\text{snoc}$ -lists.                                                           |
| $\text{segment} = \text{suffix } \text{prefix}$                                                                                   | $[A] \rightarrow [A]$                 | A contiguous stretch of $xs$ : a suffix, then a prefix of that.                                                                                                   |
| $\text{partition} = \text{concat} \circ$                                                                                          | $[A] \rightarrow [[A]]$               | This $\text{cat}$ is restricted to $[A]^* \times [A] \rightarrow [A]$ , so $ys$ is a list of non-empty segments of $xs$ .                                         |
| $\text{concat} = ([\text{nil}, \text{cat}])$                                                                                      | $[[A]] \rightarrow [A]$               | Joins the segments back up, which is why its converse splits a list.                                                                                              |
| $\text{inits}$                                                                                                                    | $[A] \rightarrow [[A]]$               | Implements $\frac{\text{prefix}}{\exists}$ , listing the prefixes by increasing length.                                                                           |
| $\text{tails}$                                                                                                                    | $[A] \rightarrow [[A]]$               | Implements $\frac{\text{suffix}}{\exists}$ by decreasing length — the opposite order.                                                                             |
| $\text{filter}(p) \triangleq \frac{\text{subseq } \text{list}(p)}{\exists} \text{ est}(R^\circ)$                                  | $[A] \rightarrow [A]$                 | The longest subsequence of $xs$ whose every element passes $p$ .<br>$\text{est}(R^\circ)$ is $\triangleq \text{est}$                                              |
| $R \triangleq \text{length} \leq \text{length}^\circ$                                                                             | $[A] \rightarrow [A]$                 | The preorder $\text{filter}$ and $\text{takewhile}$ maximise over: the longer list wins. $\geq \triangleq \leq^\circ$                                             |
| $\text{takewhile}(p) \triangleq \frac{\text{prefix } \text{list}(p)}{\exists} \text{ est}(R^\circ)$                               | $[A] \rightarrow [A]$                 | The same with $\text{prefix}$ for $\text{subseq}$ : the longest prefix whose every element passes $p$ .                                                           |
| $\text{mss} \triangleq \frac{\text{segment } \text{sum}}{\exists} \text{ est}(\geq)$                                              | $[A] \rightarrow A$                   | Maximum segment sum. $\text{segment} = \text{suffix } \text{prefix}$ splits it into $\frac{\text{prefix } \text{sum}}{\exists} \text{ est}(\geq)$ on each suffix. |

§5.8.1.  $\frac{\text{subseq}}{\exists} = ([\text{nil } \frac{1}{\exists}, (1 \times \exists) E(\text{cons}), \pi_2] \text{ cup})$

(5.8.1a)

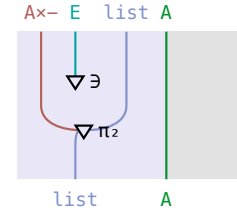
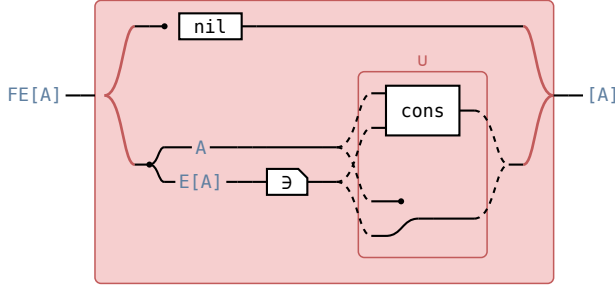
$\text{cup} \triangleq \frac{\pi_1 \exists \cup \pi_2 \exists}{\exists} : EA \times EA \rightarrow EA$ , so  $\frac{R \cup S}{\exists} = (\frac{R}{\exists}, \frac{S}{\exists}) \text{ cup}$ .

$$F(\exists)[\text{nil}, \text{cons } \cup \pi_2] = [\text{nil } \frac{1}{\exists}, (1 \times \exists)(\text{cons } \cup \pi_2)] \quad (5.8.1b)$$

the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.7.2b) at **subseq** =  $([\text{nil}, \text{cons } \cup \pi_2])$ ,  $\triangleq$  **subseq**.

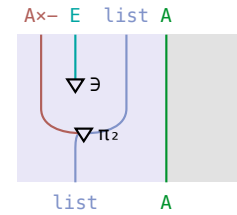
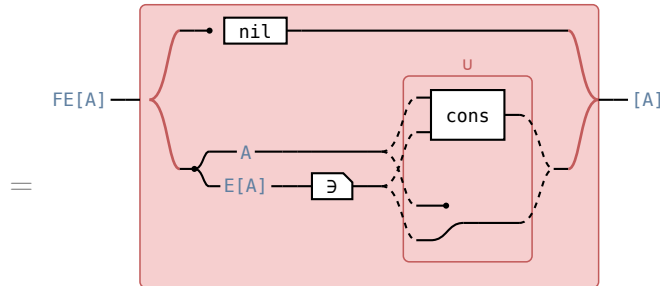
**circuit** — the fork is  $F([A]) = 1 + A \times [A]$ : **nil** above, the pair below

**Hinze–Marsden**



$F(\exists)[\text{nil}, \text{cons } \cup \pi_2]$

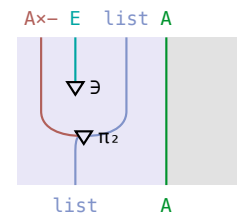
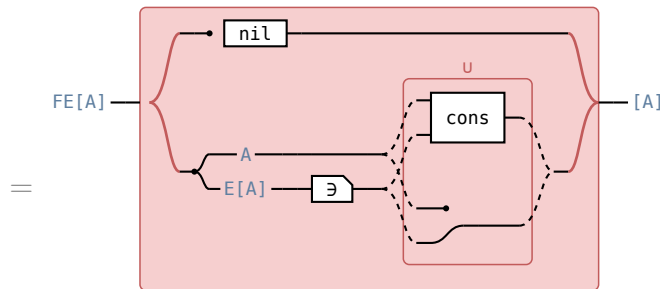
the  $\pi_2$  operand of  $\text{cons } \cup \pi_2$  under the  $1 \times \exists$  summand of  $F(\exists)$ , i.e.  $(1 \times \exists) \pi_2$



$(1+1 \times \exists)[\text{nil}, \text{cons } \cup \pi_2]$

$F(X) = 1 + A \times X \triangleq \text{subseq}$

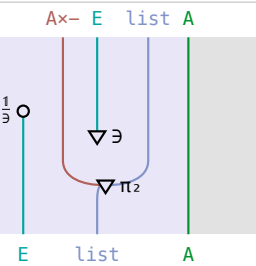
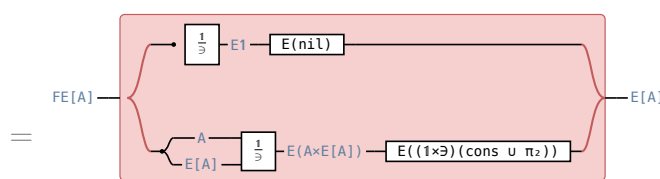
the same operand under  $1 + 1 \times \exists$ , whose  $1 \times \exists$  summand it sits in



$[\text{nil}, (1 \times \exists)(\text{cons } \cup \pi_2)]$

$R + S \triangleq [Rl, Sr], l[R, S] = R, r[R, S] = S \quad (5.3a)$

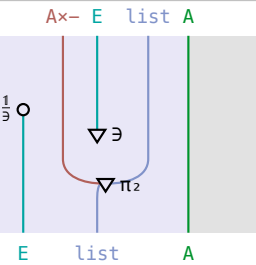
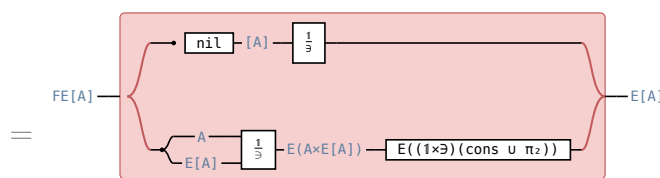
the  $\pi_2$  operand of the second arm  $(1 \times \exists)(\text{cons } \cup \pi_2)$



$[\text{nil } \frac{1}{\exists}, (1 \times \exists)(\text{cons } \cup \pi_2)]$

(5.3.1a) at  $T := [\text{nil}, (1 \times \exists)(\text{cons } \cup \pi_2)]$

the  $\pi_2$  operand under its  $1 \times \exists$ , the arm (5.8.1c)'s square rewrites,  $(1 \times \exists) \pi_2 = \pi_2 \exists$

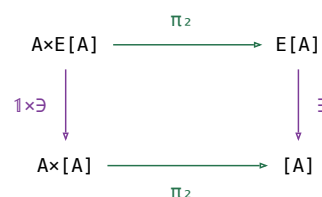


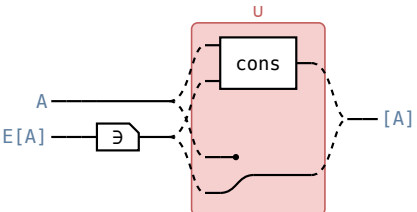
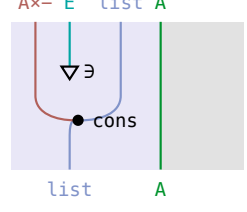
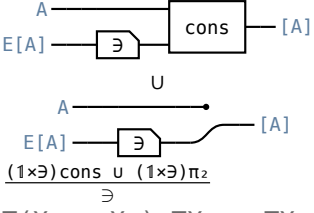
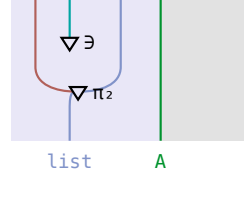
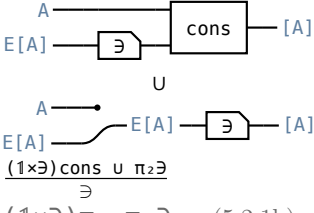
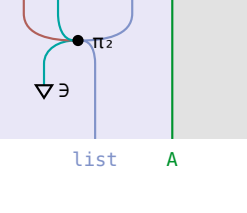
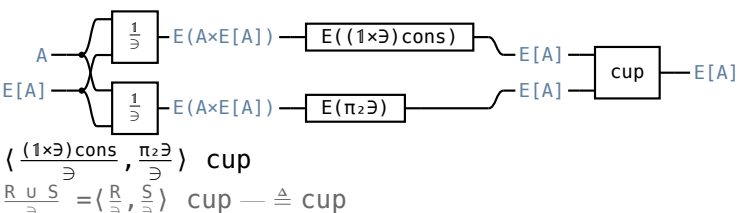
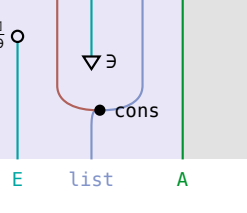
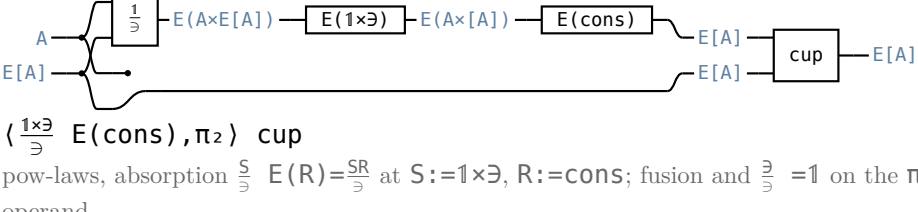
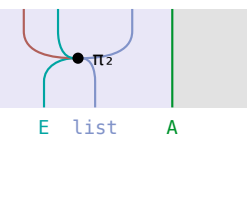
$[\text{nil } \frac{1}{\exists}, (1 \times \exists)(\text{cons } \cup \pi_2)]$

pow-laws,  $\frac{f}{\exists} = f \frac{1}{\exists}$  for  $f$  a map, at  $f := \text{nil}$

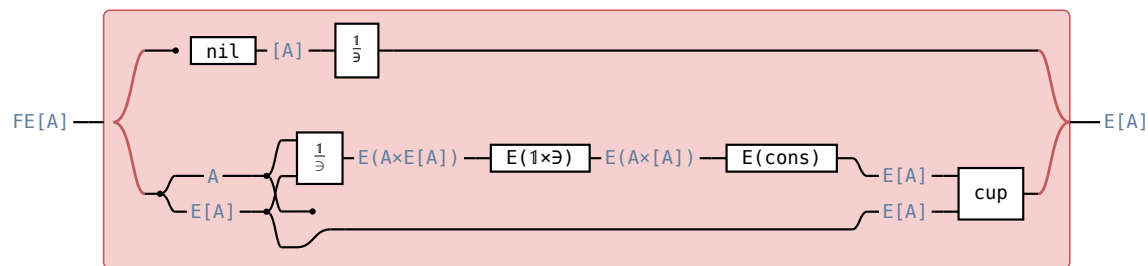
the same operand; the two rows differ only in the **nil** arm

(5.8.1c)



| $\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists} = \left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ <p>power transpose of join: the power transpose of the join of two relations is <math>\left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}</math>, where <b>cup</b> is the function that returns the union of two sets</p>                                                    | $(5.8.1d)$                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden                                                                                                                                                                                        |
|  <p><math>\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists}</math><br/>(5.8.1b)'s second branch</p>                                                                                                                                                                                                                                           |  <p>the <b>cons</b> operand of <math>\text{cons} \cup \pi_2</math></p>                                            |
|  <p><math>\frac{(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2}{\exists}</math><br/><math>T(X_1 \cup X_2) = TX_1 \cup TX_2</math> — adj-cross</p>                                                                                                                                                                                          |  <p>the <math>\pi_2</math> operand of <math>(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2</math></p> |
|  <p><math>\frac{(1 \times \exists)\text{cons} \cup \pi_2 \exists}{\exists}</math><br/><math>(1 \times \exists)\pi_2 = \pi_2 \exists</math> — (5.2.1b) at <math>\pi_2</math>, an equality because <math>1</math> is entire</p>                                                                                                                         |  <p>the <math>\pi_2</math> operand of <math>(1 \times \exists)\text{cons} \cup \pi_2 \exists</math></p>          |
|  <p><math>\left( \frac{1 \times \exists}{\exists} \text{cons}, \frac{\pi_2 \exists}{\exists} \right) \text{cup}</math><br/><math>\frac{R \cup S}{\exists} = \left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup} \triangleq \text{cup}</math></p>                                                                                          |  <p>the <b>cons</b> operand under its <math>1 \times \exists</math></p>                                         |
|  <p><math>\left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}</math><br/>pow-laws, absorption <math>\frac{S}{\exists} E(R) = \frac{SR}{\exists}</math> at <math>S := 1 \times \exists</math>, <math>R := \text{cons}</math>; fusion and <math>\frac{\exists}{\exists} = 1</math> on the <math>\pi_2</math> operand</p> |  <p>the <math>\pi_2</math> operand, bare <math>\pi_2</math></p>                                                 |

(5.8.1e)



$$\left[ \frac{\text{nil}}{\exists}, \left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \right]$$

 which writes  $P\text{cons}$ ;  $\text{cons}$  is a map, and there  $P(\text{cons}) = E(\text{cons})$  — (5.4d).

## §5.9. Lax natural transformations (LaT)

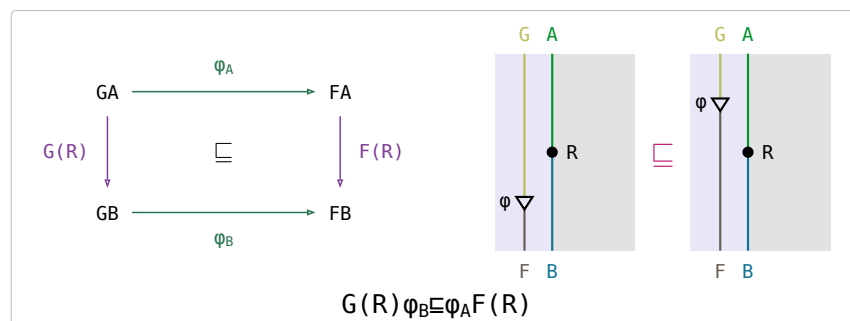
(5.9a)

For relators  $G, F : \mathcal{C} \rightarrow \mathcal{D}$  and components  $\varphi_A : GA \rightarrow FA$ ,  $\varphi$  is **lax at**  $R : A \rightarrow B$  when  $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$  — both sides  $GA \rightarrow FB$ , the left through the component at  $B$ , the right through the one at  $A$ .

$\varphi : G \Rightarrow F$  is a **lax natural transformation** (LaT) when it is lax at **every**  $R$ . (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality  $G(f)\varphi = \varphi F(f)$ : laxness is about relations only.

(5.9b)



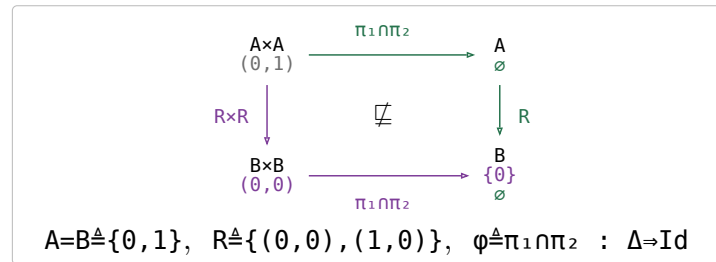
| closed under                              | commutative diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | string diagram |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| composition<br>$\psi\phi$                 | <p><math>H(R)\psi_B \sqsubseteq \psi_A G(R)</math> and <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> give <math>H(R)(\psi_B\phi_B) \sqsubseteq (\psi_A\phi_A)F(R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |
| horizontal composition<br>$\chi\circ\phi$ | <p><math>\phi : G \Rightarrow F</math> with <math>G, F : \mathcal{C} \rightarrow \mathcal{D}</math> and <math>\chi : L \Rightarrow K</math> with <math>L, K : \mathcal{D} \rightarrow \mathcal{E}</math> give <math>\chi \circ \phi : L \circ G \Rightarrow K \circ F</math>, the family <math>A \mapsto \chi_{GA} K(\phi_A)</math><br/> <math>L(G(R))\chi_{GB} \sqsubseteq \chi_{GA} K(G(R))</math> is <math>\chi</math> lax at <math>G(R)</math>; then<br/> <math>K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))</math> is <math>K</math> applied to <math>\phi</math>'s own inequation;<br/> <math>\chi := \mathbb{1}_K</math> gives <math>K(\phi) : K \circ G \Rightarrow K \circ F</math>, <math>K</math> composed on the outside; <math>\phi := \mathbb{1}_G</math> gives <math>\chi G : L \circ G \Rightarrow K \circ G</math>, <math>G</math> composed on the inside<br/> two candidates, <math>\chi_{GA} K(\phi_A)</math> and <math>L(\phi_A)\chi_{FA}</math>, ordered by <math>\sqsubseteq</math>; this row is the first</p> |                |
| union<br>$\phi \cup \psi$                 | <p><math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> and <math>G(R)\psi_B \sqsubseteq \psi_A F(R)</math> give <math>G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A)F(R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |
| a relator $K$<br>$K(\phi)$                | <p><math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> gives <math>K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |
| product<br>$\phi \times \psi$             | <p><math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi \times \psi : G \times G' \Rightarrow F \times F'</math><br/> <math>(R \times S)(U \times V) = (RU) \times (SV)</math> and monotonicity in both slots — the row above's <math>K</math> applied to the inequation, run on a bifunctor; the fork is the derived case <math>\langle \phi, \psi \rangle = \triangleleft(\phi \times \psi)</math>, and its ONE cost is the lax copy law <math>R \triangleleft \triangleleft (R \times R)</math> — rel-monoid</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |
| coproduct<br>$\phi + \psi$                | <p><math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi + \psi : G + G' \Rightarrow F + F'</math><br/> <math>(R + S)(U + V) = (RU) + (SV)</math> and monotonicity in both slots; the co-fork is the derived case <math>[\phi, \psi] = (\phi + \psi) \triangleright</math>, and <math>\triangleright</math> costs nothing<br/> the step it would need is <math>(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)</math>, the wrong direction of meet-semidistrib</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |
| meet — <b>fails</b><br>$\phi \cap \psi$   | <p>a counterexample: (5.9.1a)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |

Each row is the general slide  $R_A X \sqsubseteq X' R_B$  and  $R_B Y \sqsubseteq Y' R_C$  give  $R_A(XY) \sqsubseteq (X'Y')R_C$  at  $R_A, R_B := G(R), F(R)$  and  $X, X' := \phi_B, \phi_A$ , quantified over  $R$ ; at  $X' = X$  with  $R_A, R_B$  endorelations it is the monotonicity reading instead — the  $\forall R$  sits outside the law, and is the only difference

### §5.9.1. meet is not closed in LaT

counterexample —  $\varphi, \psi : \mathbf{G} \Rightarrow \mathbf{F}$  lax natural does NOT give  $\varphi \cap \psi$  lax natural

(5.9.1a)



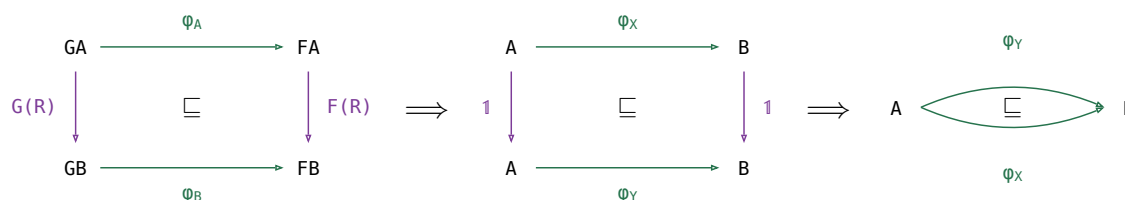
### §5.9.2. Two stacked towers

(5.9.2a)

|             | lower — inside ONE allegory      | upper — between allegories                                                                     |
|-------------|----------------------------------|------------------------------------------------------------------------------------------------|
| 0-cell      | an object $A$                    | an allegory $\mathcal{C}$ — a whole lower tower                                                |
| 1-cell      | a relation $R : A \rightarrow B$ | a relator $F : \mathcal{C} \rightarrow \mathcal{D}$                                            |
| 2-cell      | $R \sqsubseteq S$                | a LaT $\varphi : \mathbf{G} \Rightarrow \mathbf{F} \triangleq \text{lax}$                      |
| order       | $R \sqsubseteq S$                | $\varphi \sqsubseteq \psi$ componentwise                                                       |
| union       | $R \cup S$                       | $\varphi \cup \psi$ , survives (5.9c)                                                          |
| product     | $R \times S \triangleq \times$   | $\varphi \times \psi$ , survives; a TENSOR, not a categorical product; SYMMETRIC (5.2b) (5.9c) |
| coproduct   | $R + S$                          | $\varphi + \psi$ , survives; a BIPRODUCT (5.9c)                                                |
| meet        | $R \cap S$                       | $\varphi \cap \psi$ componentwise, fails (5.9.1a)                                              |
| converse    | $R^\circ$                        | $\varphi^\circ$ , fails, oplax: $\varphi_A^\circ G(R) \sqsubseteq F(R) \varphi_B^\circ$        |
| zero object | $z$ with $1 \ z = \emptyset$     | the constant relator at such a $z$ , initial and terminal                                      |

### §5.9.3. RES is a special case of LaT

(5.9.3a)



$G \triangleq \text{const } A, F \triangleq \text{const } B$  — the relators  $X \rightarrow A$  and  $X \rightarrow B$ , each sending every arrow to  $1 \triangleq \text{lax}$   
 with  $\mathcal{C}$  two objects and one non-identity arrow  $X \rightarrow Y$ , a LaT  $\text{const } A \Rightarrow \text{const } B$  is the pair  $\varphi_X, \varphi_Y : A \rightarrow B$  with  $\varphi_Y \sqsubseteq \varphi_X$  — the datum  $R \sqsubseteq S$

not every LaT is constant:  $\exists : \mathbf{P} \Rightarrow \text{Id}$  (5.4d)  $\pi_2 : (A \times -) \Rightarrow \text{Id}$  (5.8.1c)  $\triangleleft : \text{Id} = \Delta$  rel-monoid